

Università degli Studi di Padova

DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE

UNIVERSITY OF PADOVA

Department of Information Engineering

Bachelor's Degree in Computer Engineering

Final examination report

Computing the median with limited memory: study and experimentation of the multi-pass algorithm by Munro and Paterson

SUPERVISOR

Prof. Andrea Pietracaprina

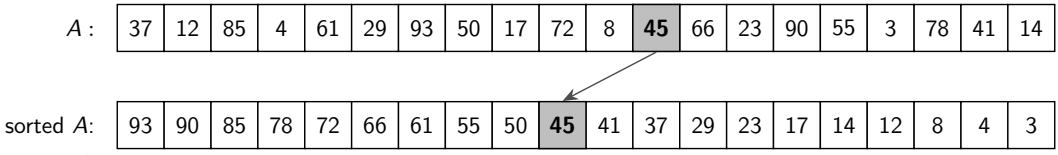
CANDIDATE

Alessandro Barbacetto

ACADEMIC YEAR 2025/2026

The selection problem

- **Input**: sequence $A = (a_1, \ldots, a_N)$ of N distinct elements from a totally ordered domain
- **Rank**: $a_{(k)}$ is the k-th **largest** element of A
- **Selection**: given $1 \leq k \leq N$, find $a_{(k)}$ without sorting the sequence
- **Median**: the case $k = \lceil N/2 \rceil$



$N = 20$, median: $k = \lceil 20/2 \rceil = 10$, hence $a_{(10)} = 45$

The SELECT algorithm

- **SELECT**: deterministic, *median of medians*
  - groups of 5 elements and the median of each group
  - pivot x: the median of the medians
  - Partition around x and recursion on the part containing the sought element
- $O(N)$ time even in the worst case, but with the whole input in memory

How can the median be found when the available memory is limited instead?

Goal of the thesis

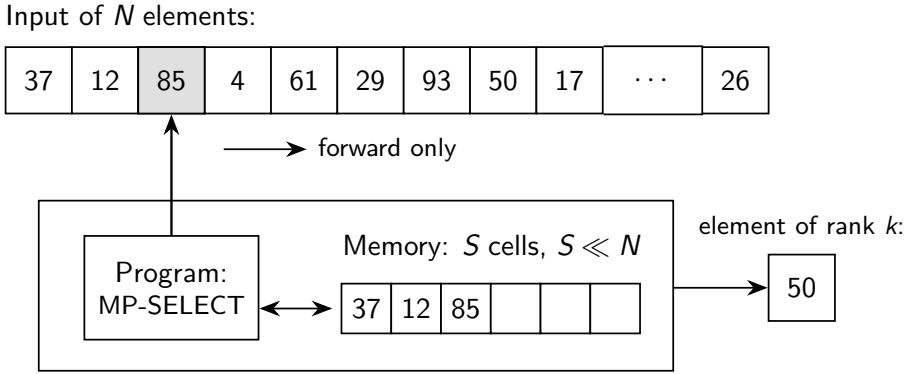
Study of the multi-pass algorithm **MP-SELECT** by Munro and Paterson for selection with limited memory, and its **experimental analysis**.

- S: available memory cells, with $S \ll N$; each cell holds one element
- P: number of **passes**, i.e. complete readings of the input; it is the cost measure
- n: **candidates**, elements lying between two filters ($f_{\min}$ and $f_{\max}$) that may still be $a_{(k)}$
- n_p: number of candidates **in pass** $p \in [1, P]$, which decreases from pass to pass

Qualitative trade-off: less memory $S \implies$ more passes P

J. I. Munro, M. S. Paterson, *Selection and sorting with limited storage*, Theoretical Computer Science 12 (1980) 315–323.

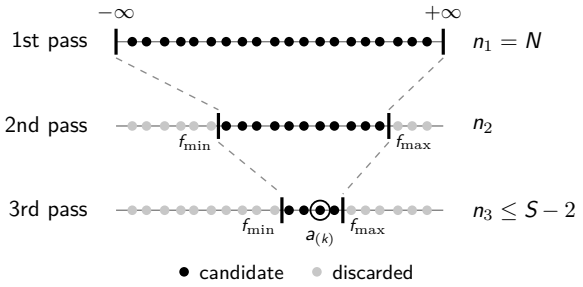
The model of computation: streaming



Streaming model (multi-pass): the input is read only sequentially, from start to end, possibly several times; at most S elements are kept in memory

RAM model (classical): the whole input is in memory, with direct access to any element

The multi-pass algorithm



In each pass:

1. **Reading**: the input is scanned once
2. **Filters**: only the candidates, between $f_{\min}$ and $f_{\max}$, are kept
3. **Sampling**: the candidates, read in sorted blocks, are halved several times (one out of two) and merged into a sample that fits in memory
4. **New filters**: from the sample, two closer filters that still enclose $a_{(k)}$

If the bound on the candidates is $\leq S - 2 \implies$ **one last pass** loads them all into memory and finds the sought element.

Goals of the experimental analysis

Bound on the candidates $\bar{n}_p$: the algorithm does not know n_p exactly, since counting it would require another pass. It therefore uses an upper bound $\bar{n}_p \geq n_p$, computed without reading the data, to:

- guarantee that the sample always fits in the S cells
- decide when to stop: if $\bar{n}_p \leq S - 2 \Rightarrow$ **last pass**

Experiments on random permutations of $1, \ldots, N$, for the median $k = \lceil N/2 \rceil$:

1. compare the number of passes P as S and N vary
2. measure how much $\bar{n}_p$ overestimates n_p
3. observe how the number of candidates evolves as the passes increase

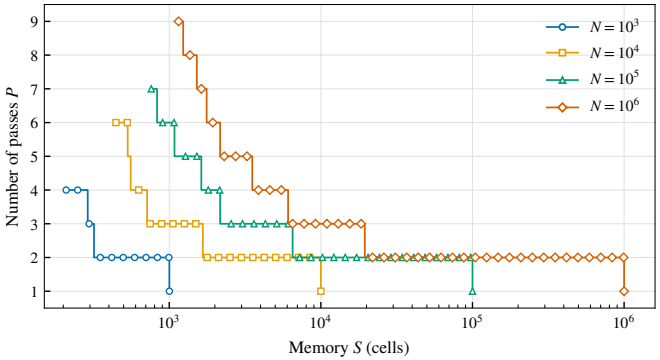
MP-SELECT algorithm implemented in **Python 3**

Results and plots produced with *Matplotlib*



https://github.com/ill-barba/bsc-thesis-unipd-median-selection

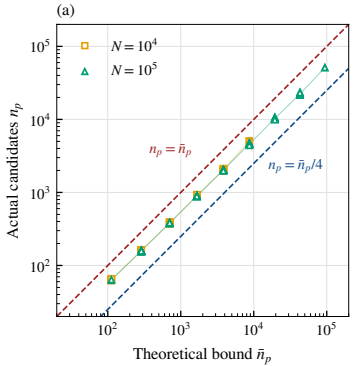
Trade-off between memory and passes

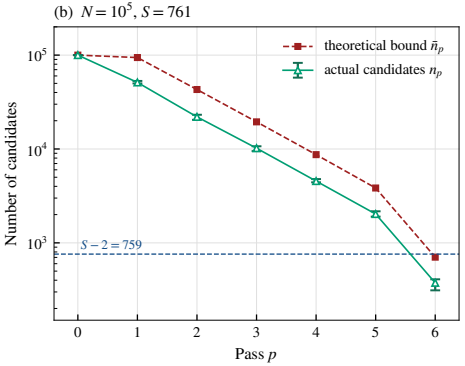


- Memory for P passes: $S \sim N^{1/P}$
- Logarithmic dependence on N, with $S = \Omega(\log^2 N)$
- Can be read both ways: $S \to P$ and $P \to S$

Theoretical bound: $\bar{n}_{p+1} = O\left(\frac{\bar{n}_p (\log \bar{n}_p)^2}{S}\right)$ **For fixed S:** $P \approx \frac{\log N}{\log S}$

Actual candidates and theoretical bound



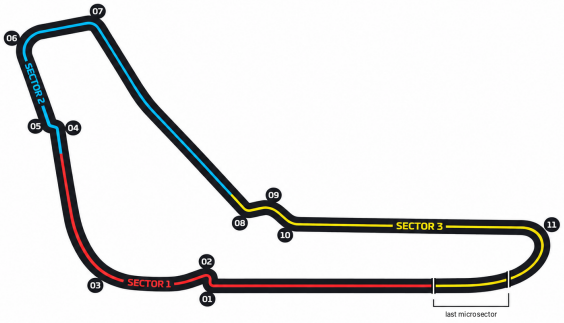


- $k = \lceil N/10 \rceil$ similar to the median; extreme ranks such as $k = 1$ have fewer candidates
- Bars initially close together due to averaging out
- "Slowdown" due to $\bar{n}_p$ driving the algorithm

Limitation: conservative bound (about twice the actual candidates), which can cost at most one more pass than necessary.

Possible development: an *ad hoc* greedy heuristic to attempt an early termination.

Identifying the thermal window



Last microsector $\sim 3.8\,\mathrm{s}$

Preparation lap j:

$$T_j = (t_{j,1}, \ldots, t_{j,N_j}), \qquad k_j = \left\lceil \frac{N_j}{2} \right\rceil$$

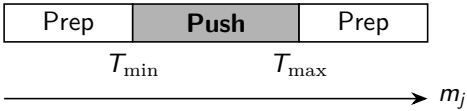
Median of the lap:

$$m_j = \text{MP-SELECT}(T_j, k_j, S)$$

First lap in the thermal window:

$$j^* = \min\{\, j \;:\; T_{\min} \le m_j \le T_{\max} \,\}$$

Graphically:



Extension to four tyres

One median for each tyre: $m_j^{FL},\ m_j^{FR},\ m_j^{RL},\ m_j^{RR}$

1. Conservative: Push only if all four are in the window

$$T_{\min} \leq m_j^w \leq T_{\max} \qquad \forall\, w \in \{FL, FR, RL, RR\}$$

2. Mean of the medians: more permissive

$$\overline{m}_j = \tfrac{1}{4}(m_j^{FL} + m_j^{FR} + m_j^{RL} + m_j^{RR}), \quad T_{\min} \leq \overline{m}_j \leq T_{\max}$$

3. Weighted mean: trade-off, different weights according to the stress on each tyre

$$\widetilde{m}_j = \alpha_{FL} m_j^{FL} + \alpha_{FR} m_j^{FR} + \alpha_{RL} m_j^{RL} + \alpha_{RR} m_j^{RR}$$
$$T_{\min} \leq \widetilde{m}_j \leq T_{\max}$$



Thermal window for each compound